

Usman Shehzad

Security Engineer | Cyber Threat Intelligence Analyst | SIEM & SOAR Specialist

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PROFESSIONAL SUMMARY

Security Engineer and Cyber Threat Intelligence (CTI) Analyst with hands-on industry experience across SIEM & SOAR operations, Continuous Threat Exposure Management (CTEM), and full-stack security platform development. Builds production-grade security systems including a CTEM platform supporting agent and agentless Vulnerability Assessment, Configuration Assessment against CIS, PCI-DSS, NIST, and CMMC frameworks, Penetration Testing workflows, and a full-featured CTI platform for IOC enrichment and threat actor tracking. Practical background deploying and managing Wazuh SIEM — including custom rule creation, decoder development, active response engineering, and agent-side automation. Certified in cyber threat intelligence and governance, with strong software engineering fundamentals across Python, FastAPI, React, and Docker.

TECHNICAL SKILLS

SIEM & SOAR: Wazuh SIEM, SIEM Rule Engineering, Custom Decoder Development, Active Response (AR), Agent-Side AR, SOAR Playbook Design, MITRE ATT&CK-Aligned Detection, Log Analysis, Elasticsearch

CTEM & VA: Continuous Threat Exposure Management, Vulnerability Assessment, Configuration Assessment, CIS Benchmarks, PCI-DSS, NIST CSF, CMMC, CVSSv4, EPSS, NVD, Penetration Testing, Agent & Agentless Scanning

Threat Intelligence: IOC Analysis, OSINT, Threat Intelligence Feeds, Threat Actor Tracking, Malware Analysis (static), Incident Response

Compliance: CIS Controls, PCI-DSS, NIST CSF, CMMC, ISO 27001, RBAC, Secure Authentication, Zero-Trust Design

Languages: Python, C#, JavaScript, C/C++, Bash

Frameworks: FastAPI, Flask, ASP.NET, React, React Native, Node.js, TailwindCSS

Infrastructure: Docker, NGINX, Apache, Elasticsearch, CI/CD (GitHub Actions), Linux

Databases: MySQL, MongoDB, SQLite3

Tools: Git, Figma, UI/UX Design

PROFESSIONAL EXPERIENCE

Security Engineer

CyberSilo (via SysByte Digital)

08/2024 – Present

onsite / Gujrat, Punjab

- Lead development and maintenance of a Continuous Threat Exposure Management (CTEM) platform — supporting agent-based and agentless Vulnerability Assessment, Configuration Assessment against CIS Benchmarks, PCI-DSS, NIST CSF, and CMMC, and integrated Penetration Testing workflows for authenticated and unauthenticated assessments.
- Built Configuration Assessment modules that automatically evaluate system configurations against framework controls, generating pass/fail findings with severity scoring and prioritized remediation guidance per compliance standard.
- Designed agent-based scanning for deep host-level VA (OS versions, installed packages, running services) and agentless scanning for restricted environments — ensuring flexible deployment across all infrastructure types.
- Deploy and manage Wazuh SIEM environments — engineering custom XML detection rules, log decoders for non-standard sources, and active response (AR) modules; build agent-side AR scripts on Linux and Windows endpoints for local containment actions (IP blocking, process kill, account disable) without manager round-trips.
- Develop MITRE ATT&CK-aligned detection rules covering lateral movement, credential access, persistence, and exfiltration TTPs — reducing false positive rates and improving SOC signal quality.
- Architected and maintain a full-featured CTI platform for IOC enrichment (IP, domain, URL, hash, email), threat actor tracking, and compliance benchmarks aligned to NIST CSF and CIS Controls — with OSINT feed enrichment, MITRE ATT&CK technique mapping, CVSSv4/EPSS-scored dashboards, multi-tenant RBAC, and Docker deployment.

Security Engineer Intern

CyberSilo (via SysByte Digital) · Internship

05/2024 – 08/2024

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- Completed a 4-month Security Engineering internship gaining hands-on experience with Wazuh SIEM — agent deployment, manager configuration, log ingestion from multiple sources, and the ruleset/decoder pipeline that transforms raw logs into structured security alerts.

- Wrote custom Wazuh XML detection rules from scratch — defining conditions, severity levels, MITRE ATT&CK technique mappings, and alert descriptions for log sources not covered by default rulesets.
- Developed custom log decoders to parse non-standard application and system log formats, enabling Wazuh to extract structured fields (user, IP, action, result) for correlation and alerting.
- Implemented active response (AR) modules on the Wazuh manager side and built agent-side AR scripts on Linux and Windows endpoints for local containment actions without central manager round-trips.
- Studied SIEM & SOAR integration patterns and gained infrastructure exposure: Elasticsearch indexing, Docker-based SIEM deployments, NGINX configuration, and Linux hardening for security tooling environments.

PROJECTS

CTEM — Continuous Threat Exposure Management Platform

2025 – Present

Stack: *Python, FastAPI, React, Docker, NGINX, NVD API, CVSSv4, EPSS, CIS / NIST CSF / PCI-DSS / CMMC*

- Built a full-featured CTEM platform (deployed at 67.202.39.233:3001) enabling enterprises to continuously discover, assess, and prioritize security exposures — covering VA, configuration compliance, and penetration testing in a single unified dashboard.
- Implemented both agent-based (deep host-level) and agentless (network-based remote) VA scanning modes — supporting environments with and without installation permissions on target hosts.
- Built Configuration Assessment modules for CIS Benchmarks, PCI-DSS, NIST CSF, and CMMC — generating per-control pass/fail findings, severity scoring, and prioritized remediation guidance per framework.
- Integrated Penetration Testing workflows for authenticated and unauthenticated assessments, validating exploitable attack paths beyond automated VA findings with active exploitation confirmation.
- Applied CVSSv4 severity scoring and EPSS exploit-likelihood prioritization across all findings; built overview dashboard with real-time exposure metrics, asset inventory, per-framework compliance posture, and CISO-ready trend reporting.
- Full infrastructure stack: FastAPI backend, React frontend, Docker, NGINX reverse proxy with TLS, and NVD API integration for continuous CVE database synchronization.

Network Vulnerability Scanning Tool

10/2025 – Present

Stack: *Python, Bash, NVD API, CVSSv4, EPSS, JSON/TXT Reporting*

- Built a standalone agentless network vulnerability scanning system for environments with restricted access and no software installation permissions — lightweight, enterprise-ready, and suitable for air-gapped environments.
- Used remote commands and network-based techniques to identify OS versions, installed packages, running services, and open ports without deploying any agent on target hosts.
- Mapped discovered versions against the NVD to detect known CVEs, automatically calculated CVSSv4 severity scores, and integrated EPSS to rank vulnerabilities by real-world exploit likelihood.
- Generated structured JSON and TXT reports for independent analysis, offline usage, and external security assessment integration.

ThreatEagle — Full-Stack Vulnerability Management Platform (Final Year Project)

10/2025 – Present

Stack: *React, FastAPI, Python, Docker, NVD API, EPSS, CVSSv4, Elasticsearch*

- Designed and built ThreatEagle as FYP — full-stack vulnerability management platform with React dashboard frontend (github.com/Usman-Shehzad/network-vulnerability-frontend) and Python/FastAPI scanning backend (github.com/Usman-Shehzad/FYP_Network_vulnerabilities_backend).
- Agentless scanning engine for OS/service/port detection, NVD CVE mapping, CVSSv4 scoring, and EPSS prioritization — with real-time React dashboard, severity heatmaps, and exportable compliance reports. Containerized via Docker for enterprise scalability.

Cyber Threat Intelligence (CTI) Platform

08/2024 – Present

Stack: *Python, FastAPI, React, Elasticsearch, SQLite3, Docker, OSINT Feeds*

- Built an enterprise-grade CTI platform for IOC enrichment (IP, domain, URL, hash, email), threat actor tracking, vulnerability monitoring, and compliance mapping against NIST CSF and CIS Controls.
- Integrated OSINT threat feeds with automated enrichment pipelines; implemented MITRE ATT&CK technique mapping, SIEM-compatible dashboards, Docker deployment with multi-tenant RBAC, and Elasticsearch-backed IOC search.

Gym Management CRM

06/2024 – 12/2025

Stack: *React, Node.js, MySQL, TailwindCSS*

- Developed a full-stack multi-tenant CRM from scratch — real-time attendance, dynamic role management, supplement e-commerce, and responsive TailwindCSS UI across multiple gym locations.

EDUCATION

Bachelor of Science — Software Engineering

PMAS-Arid Agriculture University, Rawalpindi

CGPA: 3.81 / 4.00 | Final Year Project: ThreatEagle — Full-Stack Vulnerability Management Platform

Expected 01/2026

Gujrat, Punjab

Intermediate — Computer Science

Government Zamindar Post Graduate College

Grade: B

11/2021

Gujrat, Punjab

CERTIFICATIONS

- [Certified Threat Intelligence & Governance Analyst \(CTIGA\)](#) — Red Team Leader
- [Certified Cyber Security Educator Professional \(CCEP\)](#) — Red Team Leader
- [Introduction to Cyber Security](#) — Cisco Networking Academy
- [Networking Basics](#) — Cisco Networking Academy
- [Computer Hardware Basics](#) — Cisco Networking Academy
- [Python Essentials 1](#) — Cisco Networking Academy

ACHIEVEMENTS

- Promoted to full-time Security Engineer at CyberSilo following internship performance in Wazuh SIEM rule engineering, custom decoder development, and active response automation.
- Built and deployed a live production CTEM platform (67.202.39.233:3001) supporting agent and agentless VA, CIS/PCI-DSS/NIST/CMMC configuration assessment, and penetration testing for enterprise security operations.
- Placed 2nd in college-level Designing Competition — recognized for UI/UX and visual communication skills.
- Placed 3rd in college-level Typing Competition — demonstrated precision and performance under timed conditions.